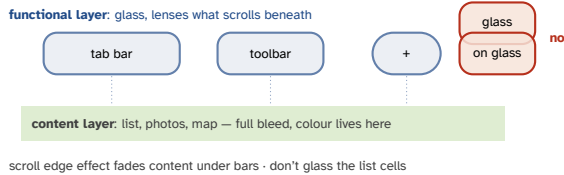
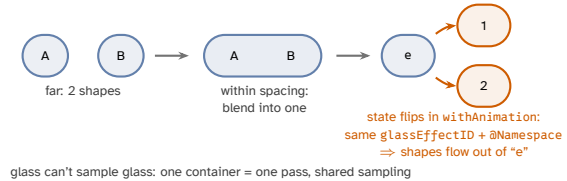


**In one line:** **Liquid Glass** is a system *material* for the **functional layer** (tab bars, toolbars, sheets, key controls) that floats *above* a full-bleed **content layer**; it lenses and adapts to what is under it. Standard components get it free with the Xcode 26 SDK; custom views opt in with `.glassEffect()`, grouped in a `GlassEffectContainer` so shapes blend and **morph**.

#### Layers (side view)



#### GlassEffectContainer(spacing: 20)



## Liquid Glass (iOS 26)

- `.glassEffect(_:in:)` — default `.regular` in a **capsule** behind the view's bounds. `.regular` = adaptive, use almost always · `.clear` = only over media, with a dimming layer · `.identity` = off, same layout. `.tint(_:)`, `.interactive()` (bounce/shimmer on touch).
- `GlassEffectContainer(spacing:)` renders its glass together; shapes nearer than spacing **blend**. Morph: `glassEffectID(_:in:) + @Namespace + state change in withAnimation`; `glassEffectUnion(id:namespace:)` joins views; `glassEffectTransition(.matchedGeometry)` or `.materialize`.
- `.buttonStyle(.glass) / .glassProminent` (primary action).
- `.tabBarMinimizeBehavior(.onScrollDown)` (iPhone) · `.tabViewBottomAccessory{}` (inline when minimised: read `tabViewBottomAccessoryPlacement`) · `Tab(role: .search) + .searchable`: the field replaces the bar.
- Toolbars share glass: `ToolbarSpacer(.fixed | .flexible)`, `.sharedBackgroundVisibility(.hidden)`, `.badge(n)`; `.scrollEdgeEffectStyle(.hard, for: .top)`.
- `.backgroundExtensionEffect()`: mirrors + blurs a view into the safe area (hero image under a sidebar). Once per screen.
- UIKit: `UIGlassEffect`, `UIButton.Configuration.glass()`. Opt-out `UIDesignRequiresCompatibility`: honoured with the 26 SDK, **ignored when building for 27**.

## Design + accessibility

- Glass = **navigation layer only**; **never glass on glass** (fills/vibrancy on top); **tint for meaning**, not decoration; drop custom bar backgrounds.
- Automatic: **Reduce Transparency** → frostier · **Increase Contrast** → black/white + border · **Reduce Motion** → no elastic effects. Test *custom* glass with each.

## Traps

- ! Glass on list cells, everywhere, or stacked on glass.
- ! Glass views without a container — no blend, no morph.
- ! “We’ll keep the old look” — opt-out ignored with the 27 SDK.
- ! `@State var m = Model() + init assignment` — error on 27.

## Remember

“Content below, glass above, one container, one ID.”

## Example

```
@Namespace private var ns
@State private var open = false
GlassEffectContainer(spacing: 16) {
  HStack(spacing: 16) {
    Button("Edit", systemImage: "pencil") {
      withAnimation { open.toggle() } }
      .buttonStyle(.glass).glassEffectID("edit", in: ns)
    if open {
      Image(systemName: "trash").padding()
        .glassEffect(.regular.tint(.red).interactive())
        .glassEffectID("trash", in: ns) // morphs out
    } }
  }
}
TabView { /* Tabs */ Tab(role: .search) { SearchView() } }
  .tabBarMinimizeBehavior(.onScrollDown)
  .tabViewBottomAccessory { NowPlayingBar() }
```

## New SwiftUI — iOS 26

- `WebView(page) + WebPage` (WebKit, `@Observable`): Load returns an `AsyncSequence` of navigation events; `callJavaScript`.
- `TextEditor(text: $attributed)` edits an `AttributedString` (rich text).
- `@Animatable` synthesises `animatableData` (`@AnimatableIgnored` opts out); a macro, so it back-deploys. Also `Chart3D`, `SwiftUI` instrument.

## New SwiftUI — iOS 27

- `@State` is a macro: an `@Observable` object is created **lazily**, **once** per view lifetime (back to iOS 17). *Source break*: default value *and* init assignment → error — drop the default (TN3211).
- `@ContentBuilder` unifies builders (faster type-checking) · documents: `WritableDocument` snapshot → `DocumentWriter` writes off-main.
- `.reorderable()` + `.reorderContainer(for:)` in any container · `.swipeActions` anywhere inside `.swipeActionsContainer()`.
- `.alert(_:item:)`, `.confirmationDialog(_:item:)` · `AsyncImage` obeys HTTP caching (`.asyncImageURLSession`).
- `Tab(role: .prominent)`, `ToolbarOverflowMenu`, `.visibilityPriority(.high)`, `.toolbarMinimizeBehavior(...)`.